

18-427/727: Board-Level RF Systems for IoT

LAB02: Antenna Design

Writeup Due: Friday, September 26th, 2025

How to submit labs:

Create a word document or PDF that answers the questions presented below. Include any screenshots or figures as requested. **Please provide a descriptive caption for each figure!**

Question:	1	2	Total
Points:	50	50	100
Score:			

Lab Outline

This lab aims to introduce the RF PCB and Antenna toolboxes in MATLAB, tools that allow you to design and simulate antennas and estimate their performance before fabricating them.

1. Half-Wavelength Dipole
 - Design and Simulation
 - Using a full-wavelength dipole
2. Patch Antenna
 - Design
 - Simulation
 - Creating Gerber files for fabrication

1. Half-Wavelength Dipole

Let's get acquainted with the MATLAB Antenna Toolbox by designing and simulating a half-wavelength dipole antenna.

```
1 dipole_radius = 1e-3;    % radius in meters
2 dipole_length = 10e-2;  % length in meters
3
4 dipole = dipoleCylindrical('Radius',dipole_radius,'Length',dipole_length);
```

This code produces a cylindrical wire antenna with radius of 1 mm and length of 10 cm. To visualize the antenna design, as well as its pattern at $f = 1$ GHz:

```
1 show(dipole)
2
3 figure
4 f = 1e9;
5 pattern(dipole,f)
```

This gets things started, but it's not quite right. See if you can adjust the antenna design a little bit to realize its ideal directivity!

5 pts

1.1 What is the directivity of this antenna at $f = 1$ GHz (this should show up in your simulation), and how does this compare to an ideal half-wavelength dipole?

5 pts

1.2 Modify the code parameters to produce an antenna pattern with the ideal half-wavelength dipole directivity at $f = 1$ GHz. Explain what change or changes you made and why.

10 pts

1.3 Provide each of the following plots of your modified dipole antenna:

1. 3D pattern
2. Elevation pattern
3. Azimuth pattern
4. S_{11} sweep that shows the center frequency of the antenna. Your sweep should be at least 200 MHz wide.
5. Impedance sweep

5 pts

1.4 What is the SWR at the center frequency of this antenna? Why isn't it 1 (i.e. why is there voltage reflection from the antenna)?

10 pts

1.5 We design dipoles to be driven at half-wavelengths. Simulate what happens if the dipole is driven at $f = 2$ GHz (i.e. the dipole is driven with a signal that equals its wavelength). Provide these plots:

1. 3D pattern
2. S_{11} sweep centered around 2 GHz
3. Impedance sweep

5 pts

1.6 What happened to the dipole's directivity at 2 GHz? What happened to its match (i.e. S_{11})?

10 pts

1.7 Why are half-wavelength dipoles preferred to full-wavelength dipoles? Please describe the standing-wave voltage and current patterns across the dipole in your answer.

2. Patch Antenna

It's Patch Time! Use the Antenna Toolbox and the RF PCB Toolbox in MATLAB to design your patch antenna and export the PCB as Gerber files, as described in the lab session. Then, provide answers to the following questions about your design.

2 pts

2.1 What is the center frequency of your design, and how does it affect the overall size of your patch?

3 pts

2.2 Provide a plot showing your antenna's geometry.

15 pts

2.3 Provide each of the following plots of your patch antenna:

1. 3D pattern
2. Elevation pattern
3. Azimuth pattern
4. S_{11} sweep that shows the center frequency of the antenna. Your sweep should be at least 200 MHz wide.
5. Impedance sweep

5 pts

2.4 Describe your design process. How many iterations of your design did you go through? How did you tweak and modify your design from the design equations?

25 pts

2.5 Submit your Gerber files for fabrication! Please include all the Gerber files, as well as the MATLAB file (.m) that generated them, in a .zip file. Each group should send the .zip file to the instructors.

Reflections

Please complete our [post-lab survey](#) for 5 points of extra credit! Attach a photo of the form completion page to your lab submission.